

## To create a Binary Search Tree using array

```
#include <stdio.h>

#define MAX 50
int bst[MAX];

// Initialize all values to -1 (empty)
void init() {
    for (int i = 0; i < MAX; i++)
        bst[i] = -1;
}

void insert(int value) {
    int index = 1;

    while (index < MAX) {
        if (bst[index] == -1) { // empty position found
            bst[index] = value;
            return;
        }
        else if (value < bst[index]) {
            index = index * 2;    // go left
        }
        else {
            index = index * 2 + 1; // go right
        }
    }
    printf("Tree is full! Cannot insert %d\n", value);
}

void inorder(int index) {
    if (index >= MAX || bst[index] == -1)
        return;

    inorder(index * 2);    // Left
    printf("%d ", bst[index]); // Root
    inorder(index * 2 + 1); // Right
}

int main() {
    int n, value;

    init();

    printf("Enter number of elements to insert: ");
    scanf("%d", &n);
```

```
printf("Enter the values:\n");
for (int i = 0; i < n; i++) {
    scanf("%d", &value);
    insert(value);
}

printf("\nInorder Traversal of BST: ");
inorder(1);
printf("\n");

return 0;
}
```